

MAE 271B Course Project

Continuous Time Kalman-Bucy Filter for GPS

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1. System Dynamics and Measurement Equations

State Dynamics (Continuous Time):

$$dx_t = Ax_t dt + G_t dW_t$$

where W_t is a standard Wiener process with diffusion Q .

Nonlinear Measurement Equation:

$$dy_t = h(x_t)dt + R^{1/2}dV_t$$

- $h(x_t)$ represents the pseudorange to satellites.
- v_t is the measurement noise (Wiener process).
- R is the measurement error covariance.

2. Linearized Kalman-Bucy Filter (KBF)

Linearization is performed about a **precomputed nominal trajectory** $\bar{x}_t$.

State Estimate Update:

$$d\hat{x}_t = f(\bar{x}_t, t)dt + K_t(dy_t - h(\bar{x}_t)dt - H_t(\hat{x}_t - \bar{x}_t)dt)$$

Gain and Covariance:

- $K_t = P_t H_t^\top R^{-1}$
- $\dot{P}_t = A_t P_t + P_t A_t^\top + Q - P_t H_t^\top R^{-1} H_t P_t$

Note: H_t and A_t are Jacobians evaluated at the nominal $\bar{x}_t$.

3. Extended Kalman-Bucy Filter (EKBF)

Linearization is performed about the **current estimate** $\hat{x}_t$.

State Estimate Update:

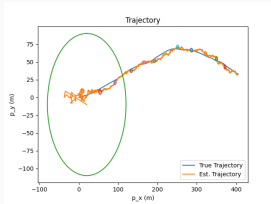
$$d\hat{x}_t = f(\hat{x}_t, t)dt + K_t(dy_t - h(\hat{x}_t)dt)$$

Gain and Covariance:

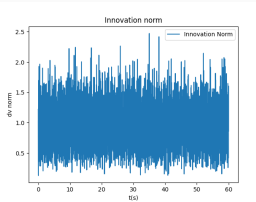
- $K_t = P_t H(\hat{x}_t)^\top R^{-1}$
- $\dot{P}_t = A(\hat{x}_t)P_t + P_t A(\hat{x}_t)^\top + Q - P_t H(\hat{x}_t)^\top R^{-1} H(\hat{x}_t) P_t$

The Jacobians are updated at every step using the latest estimate.

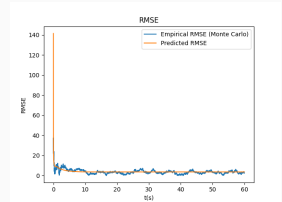
4. Experiment 1: Baseline Performance



Trajectory



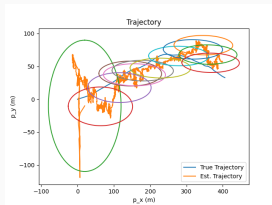
Innovation



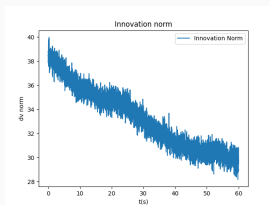
RMSE

- Well-spread satellite geometry.
- Innovation norm stable ($\approx 1.5\text{m}$).

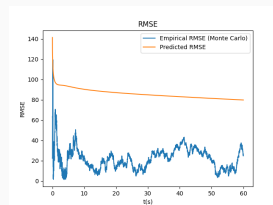
5. Experiment 2: Degenerate Geometry



Trajectory



Innovation



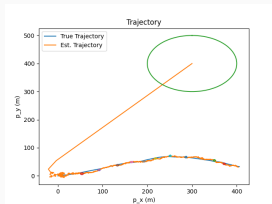
RMSE

- Satellites clustered; poor observability.
- Large uncertainty ellipses in the x-direction.

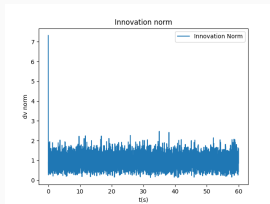
6. Baseline vs. Degenerate Geometry

Metric	Baseline	Degenerate
Geometry	Well-Spread	Clustered
Condition Number	Low ($\sim 10^1$)	High ($\sim 10^3$)
RMSE (Mean)	$< 1.0\text{m}$	$> 3.0\text{m}$
Innovation	White Noise	Biased/High Var

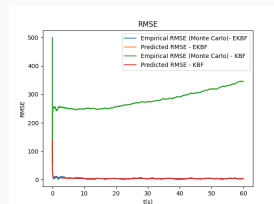
7. Filter Comparison: Robustness



EKBF Trajectory



RMSE Comparison



KBF Error

- **Initial Error:** 300–400 meters.
- **Result:** EKBF converges due to adaptive linearization. KBF fails as the nominal trajectory is invalid.